

02612 Constrained Optimization

Exercise 06

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1 Problem 1 - Inequality Constrained Quadratic Programming

Consider the QP in Example 16.4 (p.475) in Nocedal and Wright. This is an example of a convex QP in the form

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2}x'Hx + g'x \quad (1.1a)$$

$$s.t. \quad c_i(x) = a_i'x + b_i \geq 0 \quad i \in \mathcal{I} \quad (1.1b)$$

1. Make a contour plot of the problem.
2. Implement a primal active-set QP algorithm and illustrate its iterations in the contour plot for a feasible starting point
3. Implement a dual active-set QP algorithm and illustrate its iterations in the contour plot (for any starting point)
4. Implement a primal-dual interior point QP algorithm and illustrate its iterations in the contour plot (for any starting point)

2 Problem 2 - Interior-Point Algorithm for Convex Quadratic Programming

1. Write an interior-point algorithm on paper for solution of the convex quadratic program

$$\min_{x \in \mathbb{R}^n} \quad \phi = \frac{1}{2}x'Hx + g'x \quad (2.1a)$$

$$s.t. \quad A'x = b \quad (2.1b)$$

$$C'x \geq d \quad (2.1c)$$

2. Explain the Primal-Dual Interior Point Algorithm for convex QPs.
3. Implement the Primal-Dual Interior-Point Algorithm for this convex quadratic program
4. What is H , g , A , C , b , and d for the Markowitz Portfolio Optimization Problem with $R = 15$ and the presence of a risk-free security?
5. Test this algorithm on the Markowitz Portfolio Optimization Problem, i.e. compute the efficient frontier and optimal portfolio for the situation with a risk-free security. Do your algorithm give the same solution as `quadprog`?